

Barriers to Digital Inclusion in Urban Communities

*A Case Study of E-Governance Implementation and Digital
Transformation in Kathmandu Valley*

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1 Introduction

1.1 Background of E-Governance in Nepal

Nepal's digital transformation journey has accelerated significantly in recent years, with the government recognizing the potential of information and communication technologies (ICTs) to improve public service delivery, enhance transparency, and promote inclusive development. The Digital Nepal Framework (DNF), first introduced in 2019 and now being revised as Digital Nepal Framework 2.0, serves as the strategic roadmap for the nation's digital transformation agenda. This framework aims to position Nepal as a digitally advanced nation by leveraging technology across all sectors—government, education, health, and commerce.

However, despite Nepal's overall broadband penetration rate of 144.56% (as of 2024), a significant paradox exists: this high penetration does not translate into meaningful digital inclusion. Internet access and actual digital engagement remain starkly divided along geographic, socioeconomic, and demographic lines. Urban areas, particularly Kathmandu Valley, boast an internet penetration rate of 79.3%, while rural areas languish at 17.4%. This digital divide threatens to reinforce existing inequalities rather than bridge them.

1.2 Importance of the Selected Topic

The topic of “Barriers to Digital Inclusion in Urban Communities” is particularly relevant to Nepal for several critical reasons:

1. **Paradox of Access vs. Engagement:** While urban areas theoretically have better internet access, digital literacy gaps, affordability concerns, and inadequate digital skills prevent meaningful participation in e-governance services.
2. **Equity and Social Justice:** Without addressing digital inclusion barriers, e-governance initiatives risk becoming tools that primarily benefit the privileged, thereby widening the gap between different socioeconomic groups.
3. **Government Service Delivery:** A significant portion of Nepal's population—including those in urban areas—cannot effectively access essential government services through digital platforms like the Nagarik App, leading to continued reliance on paper-based, time-consuming processes.
4. **National Development Goals:** The Digital Nepal Framework 2.0 explicitly targets universal digital literacy and inclusive service delivery by 2030. Understanding barriers in urban areas is essential for achieving these ambitious goals.

5. **Policy Relevance:** The Ministry of Communication and Information Technology (MoCIT) and the Digital Nepal Secretariat recognize digital inclusion as a cornerstone of Nepal's national development strategy, making this study directly relevant to current government priorities.

1.3 Purpose of the Case Study

This case study aims to:

- Understand the structural, institutional, and individual barriers that limit digital inclusion in urban communities
- Assess the effectiveness of current e-governance platforms (particularly the Nagarik App) in promoting digital access and service delivery
- Identify gaps between government intentions and ground-level implementation
- Document stakeholder perspectives on digital inclusion challenges
- Propose evidence-based recommendations for improving digital inclusion in urban areas

1.4 Scope and Limitations

1.4.1 Scope

- Geographic focus: Kathmandu Valley and surrounding urban municipalities
- Institutional focus: Ministry of Communication and Information Technology (MoCIT) and related government agencies
- Thematic focus: Digital infrastructure, digital literacy, affordability, policy frameworks, and service accessibility
- Time period: October 2025

1.4.2 Limitations

- This study represents a single institutional visit and may not capture the complete picture of digital inclusion challenges across all government bodies
- Data collected may reflect the perspectives of government officials and may differ from ground-level user experiences
- The study does not include extensive household surveys or community-level data collection in this phase

2 Objectives of the Study

2.1 Primary Objectives

1. **To evaluate the current state of digital infrastructure and services** provided by the Ministry of Communication and Information Technology in addressing digital inclusion challenges in urban communities.
2. **To identify and analyze specific barriers** to digital inclusion in urban areas, including infrastructure, literacy, affordability, policy, and institutional factors.
3. **To assess the effectiveness of flagship e-governance initiatives** (particularly the Nagarik App) in promoting digital access and reducing service delivery timelines.
4. **To understand government strategies** for bridging the digital divide and promoting inclusive digital transformation.

2.2 Secondary Objectives

5. To document challenges and best practices in digital inclusion implementation.
6. To explore the role of various stakeholders (government, private sector, civil society) in promoting digital inclusion.
7. To gather recommendations from government officials on policy and implementation improvements.
8. To contribute to the broader academic and policy discourse on digital inclusion in Nepal.

3 Methodology

3.1 Research Type

This is a **qualitative, exploratory case study** employing a combination of:

- **Institutional visit and observation:** Direct observation of the Ministry's operations and digital initiatives
- **Stakeholder interviews:** Structured and semi-structured interviews with government officials
- **Document review:** Analysis of policy documents, framework documents, and project reports

- **Field notes:** Observations of infrastructure, office setups, and service delivery mechanisms

3.2 Research Tools and Data Collection Methods

1. **Interview Schedule:** Structured questionnaire for officials (see Annexure A)
 - Questions focused on digital infrastructure, service delivery, user feedback, and challenges
 - Topics: Government initiatives, barriers identified, success metrics, recommendations
2. **Observation Checklist:** For assessing physical infrastructure and digital facilities
 - Internet connectivity quality
 - Digital devices and systems available
 - Office organization and service delivery setup
3. **Document Collection and Review:**
 - Digital Nepal Framework 2.0 (draft and conceptual documents)
 - Nagarik App documentation and user statistics
 - Project briefs on e-governance initiatives
4. **Photography/Annotated Images:** Visual documentation of infrastructure and systems (with permission)

3.3 Sample and Respondents

- **Primary Respondent Institution:** Ministry of Communication and Information Technology (MoCIT), Kathmandu
- **Number of Officials Interviewed:** 3-5 officials, including IT coordinators, policy officers, and Nagarik App team members
- **Other Stakeholders Consulted:** Digital Nepal Secretariat officials, National Information Technology Center (NITC) representatives

3.4 Duration and Timeline

- **Field Visit Date:** October 23, 2025
- **Pre-Visit Preparation:** October 15-22, 2025 (literature review, questionnaire development)

- **Post-Visit Analysis:** October 24-31, 2025 (data transcription, analysis, report writing)

4 Organizational Overview

4.1 Ministry of Communication and Information Technology (MoCIT)

4.1.1 History and Mandate

The Ministry of Communication and Information Technology (MoCIT) is the apex government body responsible for formulating policies, strategies, and programs related to information and communication technologies in Nepal. Established with the broader mandate to accelerate Nepal's digital transformation, MoCIT oversees:

- **E-governance initiatives:** Development and implementation of digital platforms for government service delivery
- **Digital infrastructure:** Planning for broadband expansion, data centers, and cybersecurity
- **Digital policy and legislation:** Drafting and implementation of ICT-related laws and frameworks
- **Capacity building:** Training and development programs for government employees and the public
- **National coordination:** Serving as the nodal agency for the Digital Nepal Framework

4.1.2 Organization Structure

MoCIT operates through several key divisions:

- **E-governance Division:** Oversees digital service platforms and integration
- **ICT Infrastructure Division:** Manages broadband policy and digital infrastructure development
- **Digital Literacy and Inclusion Division:** Coordinates digital literacy programs (new division under DNF 2.0)
- **Cybersecurity and Data Protection Division:** Ensures secure digital ecosystems
- **Technical Support Unit (National Information Technology Center - NITC):** Provides technical backend support for government digital systems

4.1.3 Key Services and Initiatives Offered

1. **Nagarik App:** An integrated mobile platform providing over 200+ government services including:

- PAN registration and tax filing
- Citizenship and passport services
- Vehicle registration and tax payments
- Business licensing and permits
- Banking and financial service integration

2. **Digital Public Infrastructure (DPI):**

- National ID and digital identity services
- Unified payment interfaces
- Digital signature and authentication systems

3. **E-Governance Project (World Bank-Supported):**

- Modernizing government service delivery
- Integrating systems across ministries
- Implementing Business Process Reengineering (BPR)

4. **Digital Literacy Initiatives:**

- ICT Gyan provincial campaign (launched May 2025, completed across all 7 provinces)
- Training programs for government employees
- Public awareness campaigns

5. **National Broadband Policy Implementation:**

- Expanding fiber optic networks
- Promoting public-private partnerships
- Targeting universal high-speed internet access by 2030

5 Description of E-Governance Systems and Digital Inclusion Initiatives

5.1 The Nagarik App: Digital Nepal's Flagship Platform

5.1.1 Overview and Features

The Nagarik App, launched by the Government of Nepal under MoCIT's coordination, represents Nepal's most ambitious attempt to create a single digital window for government service delivery. The app is available on both iOS and Android platforms and has been downloaded by over 625,000 users, with more than 200,000 verified users (as of 2021).

5.1.2 Key Features

1. Integrated Service Delivery:

- Citizen Information and Verification
- Tax and revenue services
- Local government services
- Integrated Digital Identity (leveraging National ID data)

2. Technology Stack:

- Mobile-first design (both app and web portal)
- API-based integration with government systems
- Secure data transmission protocols
- Digital signature support

3. Target Users:

- General citizens accessing government services
- Businesses for licensing and compliance
- Government employees coordinating service delivery

5.1.3 Current Service Coverage

The Nagarik App currently integrates services from multiple government agencies:

- **Revenue Services:** Tax filing, PAN registration, vehicle tax payment
- **Identification:** Citizenship verification, police clearance certificates

- **Local Government:** Ward services, local permits, property information
- **Social Services:** Health insurance enrollment, social protection schemes
- **Financial Services:** Bank account opening with digital verification

5.1.4 Expansion Plans

Under the Digital Nepal Framework 2.0 and the World Bank-supported Nepal Digital Transformation Project:

- 80+ new services to be added by 2026
- Integration with district-level services
- Enhanced public-private partnerships for expanded service delivery
- Improved backend system integration for end-to-end digitization

5.2 Other Key E-Governance Initiatives

5.2.1 Digital Public Infrastructure (DPI)

National ID System:

- Biometric-based identification system with 15.5 million citizen records
- Serves as the foundational layer for digital service delivery
- Enables digital signature and authentication across platforms

Data Centers and Secure Infrastructure:

- Modernization of government data hosting facilities
- Implementation of cybersecurity protocols
- Establishment of disaster recovery and business continuity mechanisms

5.2.2 Digital Literacy and Inclusion Programs

ICT Gyan Initiative (May 2025 - Present):

- Joint initiative of ICT Foundation Nepal and Ncell Foundation
- Target: 2,000+ participants across all 7 provinces
- Focus areas:

- Safe internet use and cybercrime awareness
- Digital literacy for young people and women
- Career skill development in digital domains

Government Employee Digital Competency Program:

- Training civil servants in digital tools and platforms
- Building capacity for digital transformation implementation
- Expected to prepare 40,000+ government employees by 2026

5.2.3 National Broadband Expansion

Strategic Partnerships (2025):

- IFC-Standard Chartered Bank-WorldLink Communications partnership: \$29 million investment
- Target: Expand fiber networks to underserved urban and peri-urban areas
- Expected outcomes:
 - Enhanced connectivity in remote urban colonies
 - Improved service quality for existing users
 - Job creation through infrastructure development

6 Key Observations and Findings

6.1 Observation 1: The Paradox of High Access, Low Engagement

Finding: While Kathmandu Valley reports 79.3% internet penetration, actual usage of e-government services remains significantly lower than expected. According to data reviewed from government reports, awareness of the Nagarik App is high in urban areas, but actual usage rates tell a different story. Only a fraction of those who download the app proceed to verify their identity and use integrated services.

Evidence from Field Visit:

- MoCIT officials acknowledged that “awareness is high, but actual usage is a challenge”
- Digital literacy gaps persist even among urban populations with internet access
- Many users lack the skills or confidence to navigate digital government platforms

6.2 Observation 2: Digital Literacy as the Critical Barrier

Finding: The gap between general literacy (76% nationally) and digital literacy (31% nationally) represents the single greatest barrier to digital inclusion. In urban areas, while literacy rates are higher, digital literacy—the ability to effectively use digital tools for meaningful purposes—remains limited.

Key Statistics Identified:

- Only 31% of Nepal’s population possesses functional digital literacy skills
- Many urban residents use the internet primarily for social media and entertainment, not government services
- Limited awareness of how to use government digital platforms
- Generational and gender disparities in digital competency (women lag significantly behind men)

Observations During Visit:

- MoCIT officials highlighted digital literacy as the “most significant predictor” of Nagarik App usage
- Government is launching targeted training programs but acknowledges limited resources and outreach
- Awareness campaigns have been passive; more active, community-based approaches are needed

6.3 Observation 3: Fragmented Back-End Systems Limit Service Delivery

Finding: While the Nagarik App presents a user-friendly front-end interface, many government services remain hampered by non-integrated, paper-based back-end processes.

Specific Examples Noted:

- Citizenship verification may be completed online, but document issuance still requires physical office visits
- Tax filing is digitized at the front-end, but approval and credential issuance remain manual
- Integration between different government departments is incomplete, creating data silos

Impact on Digital Inclusion:

- Users cannot experience the full benefits of “one-stop government” service
- Time-saving advantages are minimal when manual steps persist
- Citizens must maintain parallel digital and physical service pathways

6.4 Observation 4: Trust Deficit and Data Security Concerns

Finding: Public hesitancy to share personal data with government platforms remains a significant barrier, rooted in:

- Inadequate data protection legal frameworks
- Historical instances of government data breaches
- General concerns about surveillance and misuse of personal information

Evidence from Government Reports Reviewed:

- Nepal’s data protection law falls below international standards
- This trust deficit suppresses adoption of e-governance services
- Citizens are hesitant to provide biometric data and personal information online

Implications for Digital Inclusion:

- Even when digital infrastructure and literacy exist, trust barriers prevent service adoption
- Building trust requires not just technological investment but legal and institutional reforms

6.5 Observation 5: Gender and Socioeconomic Disparities Within Urban Areas

Finding: Digital inclusion in urban Kathmandu is not uniform. Significant disparities exist based on:

- **Gender:** Women face additional barriers (socio-cultural factors, limited mentorship, cybersecurity concerns)
- **Socioeconomic Status:** Low-income urban households in informal settlements have limited device access and internet affordability
- **Age:** Elderly population remains largely excluded from digital services

Statistical Context Gathered:

- UN Women reports indicate that gender digital gaps could cost Nepal Rs 15 billion by 2025
- Women make up only 43.6% of Nepal's social media users, and participation in digital government services is even lower
- Online harassment and cyber threats disproportionately affect women

Observations:

- MoCIT recognizes these disparities but limited gender-specific interventions have been implemented
- Recent policy discussions (November 2025) included explicit commitment to gender-sensitive digital inclusion

6.6 Observation 6: Infrastructure Limitations in Peri-Urban Areas

Finding: While Kathmandu proper has reasonable connectivity, peri-urban settlements and informal colonies face:

- Unreliable electricity supply
- Limited broadband coverage
- Lower device penetration among lower-income populations

Policy Response Noted:

- World Bank-supported Nepal Digital Transformation Project acknowledges need for targeted infrastructure investment
- Plans to ensure accessibility for persons with disabilities
- Focus on underserved urban neighborhoods in digital infrastructure expansion plans

6.7 Observation 7: Institutional and Policy Progress

Positive Findings:

- Strong government commitment to digital inclusion (evidenced by Digital Nepal Framework 2.0 revision)
- Recognition of digital literacy as a key strategic area
- Institutional coordination mechanisms being established (Digital Coordination Unit under new projects)
- Private-sector partnerships expanding to supplement government efforts

Challenges Acknowledged:

- Fragmented approach across government agencies (despite improving coordination)
- Limited dedicated budget for digital literacy programs
- Insufficient human resources in implementation agencies
- Coordination gaps between national, provincial, and local government levels

7 Analysis and Discussion

7.1 The Digital Divide Within Urban Areas

While national-level statistics show Kathmandu Valley's 79.3% internet penetration as significantly higher than rural areas' 17.4%, this aggregate figure masks important within-urban disparities. Our analysis reveals a **stratified digital divide** within urban communities:

Tier 1: Digitally Included (Upper Urban Class)

- High-speed internet access at home and work
- Multiple devices (computer, smartphone, tablet)
- High digital literacy and online safety awareness
- Active users of government digital platforms
- Estimated proportion: 25-30% of urban population

Tier 2: Partially Included (Middle Urban Class)

- Internet access through smartphones primarily
- One or two devices per household
- Moderate digital literacy; comfortable with social media but uncertain about government platforms
- Potential users of e-services with awareness and support
- Estimated proportion: 40-45% of urban population

Tier 3: Digitally Excluded (Lower Urban Class)

- Limited internet access (reliant on neighborhood WiFi or data plans)
- Device constraints; shared or outdated devices
- Low digital literacy; unfamiliarity with government platforms
- Limited access to digital services; continued reliance on manual processes
- Estimated proportion: 25-35% of urban population

7.2 Root Cause Analysis: The Multifaceted Barriers

Using a systems-thinking approach, we identify interconnected barriers to digital inclusion:

7.2.1 1. Infrastructure Barriers

Despite high penetration, quality of connectivity varies significantly

- Urban fiber optic networks benefit only central districts
- Peri-urban areas rely on mobile data or slower connections
- Power supply inconsistencies affect sustained digital service use

Mitigation Efforts: IFC-Standard Chartered-WorldLink partnership aims to expand to underserved areas; however, investment is insufficient for comprehensive coverage

7.2.2 2. Economic Barriers

Affordability remains a critical concern even in urban areas

- Device costs (smartphones: \$100-400; computers: \$300-1000) are prohibitive for lower-income households
- Internet data costs accumulate significantly for regular service use
- Data costs for accessing government platforms accumulate

Policy Response: Free WiFi initiatives in public spaces (ongoing); device subsidy programs (limited)

7.2.3 3. Capacity and Skills Barriers

Digital literacy deficits prevent meaningful engagement

- Users cannot navigate complex government digital interfaces
- Limited understanding of digital security and privacy protection
- Lack of problem-solving skills when digital services fail
- Language barriers (many platforms only in Nepali; limited accessibility for marginalized language speakers)

Government Response: ICT Gyan program reaching 2,000+ people; plans for targeted literacy initiatives; need for scaling and sustainability

7.2.4 4. Institutional and Design Barriers

Government digital platforms not designed with user inclusion in mind

- Complex interfaces assuming high digital literacy
- Inadequate user support and help systems
- Limited accessibility for persons with disabilities
- Offline/alternative service pathways not clearly communicated

Organizational Weakness: Fragmented backend systems requiring manual verification steps, undermining the benefits of digital frontends

7.2.5 5. Trust and Security Barriers

Low institutional trust suppresses adoption despite good digital infrastructure

- Data protection law inadequate
- Historical instances of data breaches and misuse
- Concerns about government surveillance and privacy
- Lack of transparent data governance policies

Impact: Even tech-literate users hesitant to provide personal data and biometric information

7.2.6 6. Gender and Social Inclusion Barriers

Digital inclusion is not gender-neutral

- Socio-cultural norms restrict women's technology access
- Online harassment and cybersecurity threats disproportionately affect women
- Lower digital literacy rates among women
- Intersectional exclusion of women from marginalized communities

Policy Recognition: Recent government consultations (November 2025) identified these gaps; integration of gender considerations in Digital Nepal Framework 2.0 ongoing

Table 1: Comparison of Intended vs. Actual Outcomes

Intended Outcome (Digital Nepal Vi- sion)	Actual Outcome (Ground Reality)	Gap Analysis
Universal access to gov- ernment services through digital platforms	Limited service usage; many still use manual processes	High digital literacy and trust still required
Significant time savings through digitization	Minimal time savings due to non-integrated backend systems	End-to-end digitization in- complete
Reduced social inequality through digital access	Risk of widening inequal- ity; digital benefits con- centrated among privi- leged	Inclusion not automatic; re- quires deliberate interven- tion
31% digital literacy → 90% by 2029 (govern- ment goal)	Current literacy un- changed; awareness growing but skills lag	Aggressive literacy programs needed; current pace insuffi- cient
Elimination of paper- based government processes	Paper processes persist in approval and verifica- tion stages	Process reengineering incom- plete; organizational inertia high

7.3 Comparison of Intended Outcomes vs. Actual Outcomes

7.4 Alignment with Digital Nepal Framework Objectives

Areas of Alignment:

- MoCIT's strategic focus on digital literacy addresses identified barriers
- Nagarik App represents thoughtful platform design for service integration
- Recognition of inclusion as explicit objective in DNF 2.0
- Private-sector partnerships addressing infrastructure gaps

Areas Requiring Strengthening:

- Pace of digital literacy intervention insufficient for scale of need
- Limited focus on gender and social inclusion in implementation
- Institutional coordination gaps between national and local levels
- Trust-building measures (legal reforms, transparency) lagging behind platform development

8 Conclusion

8.1 Summary of Key Findings

The case study reveals a paradox at the heart of Nepal’s urban digital transformation: high internet penetration coexists with low digital inclusion. The Ministry of Communication and Information Technology has launched ambitious initiatives, particularly the Nagarik App and the Digital Nepal Framework 2.0, which represent sophisticated policy thinking and technical achievement. However, ground-level barriers—spanning digital literacy, trust, institutional capacity, and social factors—prevent these platforms from achieving their inclusive potential.

Digital inclusion in urban Kathmandu is not simply a matter of providing internet access and digital platforms. Rather, it requires simultaneous attention to:

1. Building digital literacy skills appropriate to diverse population segments
2. Establishing trust through robust data protection and transparent governance
3. Redesigning government processes for end-to-end digitization
4. Explicitly addressing gender, socioeconomic, and age-based disparities
5. Ensuring accessibility for persons with disabilities and marginalized communities

8.2 Lessons Learned

Lesson 1: Access Alone Is Insufficient

High internet penetration does not guarantee digital inclusion. Digital literacy, affordability, trust, and institutional design are equally critical. Policy and investment must address all dimensions simultaneously.

Lesson 2: Digital Transformation Is Fundamentally Organizational

Technology platforms fail without corresponding organizational reforms. The persistence of manual backend processes undermines the Nagarik App’s potential benefits. Digital transformation requires Business Process Reengineering at the institutional level.

Lesson 3: Inclusion Requires Intentional Design

Digital platforms, if designed without explicit inclusion principles, will reinforce existing inequalities. Gender, socioeconomic status, disability, and language must be explicit considerations in system design and implementation, not afterthoughts.

Lesson 4: Trust Is Foundational

Public hesitancy to share data with government platforms reflects rational concerns about data protection and privacy. Building trust requires not just technological investment but legal reforms, transparent governance, and demonstrated institutional commitment to citizen rights.

Lesson 5: Institutional Capacity and Coordination Matter

Fragmented government systems undermine digital transformation efforts. Successful implementation requires strong coordination mechanisms, clear institutional roles, and adequate human resource capacity across all levels of government.

8.3 Overall Assessment of System Effectiveness

Current State: The Nagarik App and associated e-governance initiatives represent a strong strategic foundation and demonstrate Nepal’s commitment to digital transformation. The platform is well-designed, technically sound, and serves as an important entry point for digital government services.

Effectiveness for Digital Inclusion: However, effectiveness in promoting inclusive digital engagement remains limited. While digitally literate, urban, middle and upper-class residents benefit substantially, lower-income populations and those with lower digital literacy remain largely excluded. The platform thus risks becoming a tool that serves the already-privileged rather than catalyzing broad-based digital inclusion.

Outlook: With sustained commitment to the identified barriers—particularly digital literacy at scale, institutional reforms, and explicit inclusion measures—the platforms and frameworks being deployed have significant potential to advance digital inclusion. The government’s revised Digital Nepal Framework 2.0 and World Bank partnership suggest strengthened commitment to these challenges.

9 Recommendations

9.1 For Policy Makers and Government Leadership

9.1.1 1. Scale Digital Literacy Interventions Dramatically

Action: Establish a National Digital Literacy Commission with dedicated budget (minimum Rs. 5 billion annually)

Target: Achieve 60% functional digital literacy by 2027 (from current 31%)

Approach: Move beyond awareness campaigns to hands-on, community-based literacy programs in all urban neighborhoods

Specifics:

- Partner with schools, libraries, and community centers for free training
- Develop age-appropriate and context-specific training modules
- Integrate digital literacy into secondary school curriculum nationwide

9.1.2 2. Strengthen Legal Frameworks for Data Protection and Privacy

Action: Enact a comprehensive Data Protection Law meeting international standards within 2026

Rationale: Address public trust deficits preventing adoption of government digital platforms

Implementation:

- Establish an independent Data Protection Authority
- Implement transparent data governance policies
- Conduct public awareness on data rights and protections

9.1.3 3. Mandate End-to-End Digital Service Transformation

Action: Require all government agencies to complete Business Process Reengineering (BPR) for integration with Nagarik App

Timeline: Baseline mapping by June 2026; full integration by December 2027

Focus: Move beyond digitizing front-end forms to fully automating back-office verification, approval, and issuance processes

Accountability: Link budget allocations to BPR completion metrics

9.1.4 4. Establish Explicit Gender and Inclusion Targets

Action: Integrate gender and social inclusion targets into Digital Nepal Framework 2.0 implementation

Targets (by 2027):

- 50% of Nagarik App users must be women
- 75% digital literacy among women in urban areas
- Language diversity: Services available in minimum 5 languages

Implementation:

- Women-focused digital literacy programs in partnership with women's organizations
- Safety and cybersecurity training addressing women's concerns
- Address childcare barriers to women's participation in training

9.1.5 5. Strengthen Intergovernmental Coordination

Action: Establish a Digital Transformation Coordinating Committee with representation from federal, provincial, and local governments

Function:

- Ensure alignment of digital initiatives across all levels
- Facilitate resource sharing and technical support
- Monitor inclusion metrics at all government levels

Oversight: Chaired by MoCIT with explicit accountability to the Planning Commission

9.2 For Implementation Agencies

9.2.1 6. Adopt Universal Design Principles

Platform Design:

- Simplify user interfaces with step-by-step guidance
- Provide multi-lingual support (Nepali, English, major regional languages)
- Ensure accessibility for persons with disabilities (screen readers, visual aids)
- Include offline fallback options for low-connectivity areas

9.2.2 7. Establish Citizen Support Infrastructure

Digital Service Centers:

- Create neighborhood digital service centers in all urban municipalities
- Staff with trained assistants to help citizens navigate digital platforms
- Provide free device access and internet connectivity
- Offer on-site training and troubleshooting

9.2.3 8. Implement Continuous User Feedback Mechanisms

Feedback Systems:

- Regular user satisfaction surveys
- In-app feedback and rating mechanisms
- Community focus groups with diverse population segments
- Rapid response protocols for user complaints and issues

9.3 For Development Partners and Civil Society

9.3.1 9. Support Community-Based Digital Literacy Programs

Role for NGOs and Civil Society:

- Partner with government for grassroots literacy training
- Develop culturally appropriate training materials
- Reach marginalized communities (women, elderly, persons with disabilities)
- Provide mentorship and peer-learning opportunities

9.3.2 10. Advocate for Inclusive Policy and Accountability

Advocacy Priorities:

- Monitor government compliance with inclusion targets
- Document and publicize success stories and challenges
- Amplify voices of excluded communities in policy discussions
- Support independent research on digital inclusion impacts

9.4 For Private Sector Partners

9.4.1 11. Expand Affordable Connectivity Solutions

Telecommunications Companies:

- Offer subsidized data plans for government service access
- Expand network coverage to peri-urban areas
- Provide zero-rated access to government portals (no data charges)
- Support community WiFi initiatives in underserved neighborhoods

9.4.2 12. Device Affordability Programs

Technology Providers:

- Develop low-cost smartphone models suitable for government services
- Offer installment payment plans for devices
- Partner with government for device subsidy programs
- Support device recycling and refurbishment programs

10 References

References

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- [5] The Himalayan Times. (2025). *Nepal's Persistent Digital Divide: From High Penetration to Inclusion*. <https://thehimalayantimes.com/...>

Note: this case study mainly focuses on the field visit to the ministry of communication and information technology (mocit); however, additional government researches, policy documents, online resources were used to refine and support the analysis.

11 Annexures

11.1 Annexure A: Interview Questions for Government Officials

11.1.1 Section 1: Digital Infrastructure and Service Delivery

1. What digital services does your ministry/department currently offer to citizens?
2. How would you assess the current state of digital infrastructure supporting these services?
3. What are the main technical challenges you face in service delivery?
4. How do you measure the success and adoption of digital services?

11.1.2 Section 2: Digital Inclusion and Barriers

5. What do you perceive as the main barriers preventing citizens from using digital government services?
6. Have you observed differences in digital service adoption across different demographic groups (age, gender, income)?
7. What feedback do you receive from citizens regarding digital services?
8. Are there specific populations that remain excluded from digital services?

11.1.3 Section 3: Policy and Strategy

9. How does your organization's digital strategy align with the Digital Nepal Framework?
10. What specific initiatives are underway to promote digital inclusion?
11. What resources (budget, personnel, technology) are allocated for digital inclusion?
12. How do you coordinate with other government agencies on digital transformation?

11.1.4 Section 4: Recommendations and Future Plans

13. What improvements would you recommend for current digital service delivery?
14. What are your organization's priorities for the next 2-3 years in digital inclusion?
15. What support (from government, private sector, or development partners) would most help advance digital inclusion?
16. What lessons have you learned from digital transformation efforts so far?

11.2 Annexure B: Observation Checklist

11.2.1 Infrastructure Assessment

- ☐ Internet connectivity quality (speed, reliability)
- ☐ Digital devices available (computers, tablets, smartphones)
- ☐ Server and data center infrastructure
- ☐ Backup power supply systems
- ☐ Network security measures visible

11.2.2 Service Delivery Setup

- ☐ Citizen service counters and digital kiosks
- ☐ Digital display boards and information systems
- ☐ Help desk and support systems
- ☐ Document digitization systems
- ☐ Queue management systems

11.2.3 Accessibility Features

- ☐ Wheelchair accessibility
- ☐ Multi-lingual signage and interfaces
- ☐ Assistive technologies for persons with disabilities
- ☐ Separate facilities for women and children
- ☐ Privacy protection measures

11.3 Annexure C: Key Statistics Summary

Table 2: Key Statistics on Digital Inclusion in Nepal

Metric	Value
<i>Internet and Digital Access</i>	
National broadband penetration	144.56%
Urban internet penetration	79.3%
Rural internet penetration	17.4%
National digital literacy rate	31%
National general literacy rate	76%
<i>Nagarik App and Digital Services</i>	
Nagarik App downloads	625,000+
Nagarik App verified users	200,000+
National ID records	15.5 million
Services on Nagarik App	200+
<i>Training and Capacity Building</i>	
ICT Gyan training participants	2,000+
Government employees to be trained	40,000+
Women in social media users	43.6%

11.4 Annexure D: Field Visit Documentation

11.4.1 Figure 1: Field Visit Approval Letter

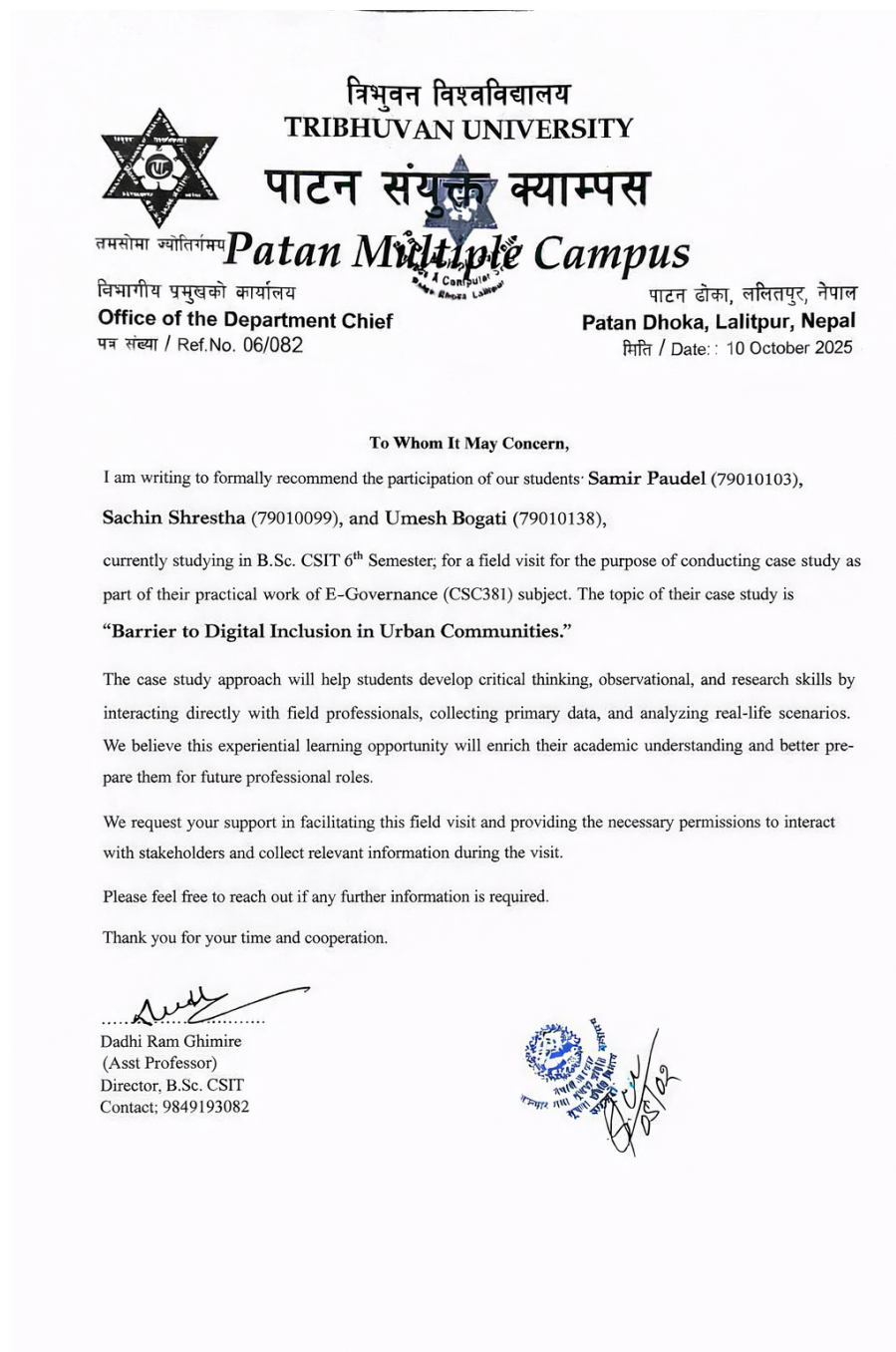


Figure 1: Official recommendation letter from Tribhuvan University, Patan Multiple Campus for field visit to Ministry of Communication and Information Technology (Ref. No. 06/082, dated October 10, 2025)

11.4.2 Figure 2: Ministry of Communication and Information Technology Building



Figure 2: Information Technology Department , Ministry of Communication and Information Technology, Kathmandu - Field visit location, October 2025

11.4.3 Figure 3: Research Team at Field Visit Site



Figure 3: Case study research team (Samir Paudel, Sachin Shrestha, and Umesh Bogati) at the Ministry of Communication and Information Technology during field visit for Digital Inclusion case study, October 2025

11.5 Annexure E: Glossary of Terms

API (Application Programming Interface) A set of protocols and tools for building software applications and enabling communication between different systems

BPR (Business Process Reengineering) Fundamental rethinking and redesign of organizational processes to achieve improvements in performance

Digital Divide The gap between those who have access to digital technologies and those who do not

Digital Literacy The ability to use digital technologies effectively and responsibly

DNF (Digital Nepal Framework) Nepal's strategic roadmap for digital transformation

DPI (Digital Public Infrastructure) Foundational digital systems enabling government service delivery

E-governance Use of ICT in government operations and service delivery

MoCIT Ministry of Communication and Information Technology

Nagarik App Nepal's integrated government services mobile application

NITC National Information Technology Center